

TOPS TECH Task 1 (CS)

Task -1 Types of Malwares Viruses

1. **Create a table listing the main differences between viruses, worms, ransomware, Trojans, spyware, and adware, and give a real-world app or website example where each type of malware could potentially cause harm.**

Ans:

Comparison of Different Types of Malware

Objective

- The objective of this task is to understand the different types of malware, their working mechanisms, methods of spreading, primary objectives, effects on computer systems, and real-world examples where each type of malware can potentially cause harm.
- Understanding these malware types helps users and organizations implement effective cybersecurity measures to protect their systems and sensitive information.

Comparison of Different Types of Malware :

Malware Type	Definition	Method of Spread	Primary Purpose	Effects on the System	Real-World Example
Virus	A virus is a malicious program that attaches itself to legitimate files or programs and executes when the infected file is opened by the user.	Infected files, email attachments, USB drives, software downloads.	To damage or modify files and disrupt normal system operations.	Corrupts files, slows system performance, causes crashes, and spreads to other files.	Microsoft Word Documents received through email attachments may contain macro viruses that infect a user's computer.

TOPS TECH Task 1 (CS)

Worm	A worm is a standalone malware that can replicate itself automatically without requiring user interaction.	Computer networks, operating system vulnerabilities, internet connections.	To spread rapidly across networks and consume system resources.	Increases network traffic, slows computers, and infects multiple systems automatically.	Corporate Office Network where a worm can spread from one computer to all connected devices.
Ransomware	Ransomware is malware that encrypts files or locks a system and demands payment from the victim to restore access.	Phishing emails, malicious websites, fake software updates, exploit kits.	To extort money from victims by denying access to important data.	Encrypts files, displays ransom notes, and makes important documents inaccessible.	Hospital Management System where patient records can become inaccessible due to encrypted files.
Trojan Horse	A Trojan Horse is malware that disguises itself as legitimate software to trick users into installing it.	Fake software, cracked applications, phishing emails, malicious downloads.	To steal information, create backdoors, or install additional malware.	Provides attackers with unauthorized access and compromises system security.	Fake Google Chrome Installer downloaded from an unofficial website.
Spyware	Spyware is malicious software designed to secretly monitor user activities and collect confidential information without the user's knowledge.	Free software bundles, malicious browser extensions, infected websites, fake applications.	To steal passwords, banking information, browsing history, and personal data.	Records keystrokes, monitors online activity, and sends sensitive information to attackers.	Online Banking Website where login credentials and financial information can be stolen.

TOPS TECH Task 1 (CS)

Adware	Adware is software that automatically displays unwanted advertisements and redirects users to advertising websites.	Bundled freeware, browser extensions, and suspicious software installations.	To generate advertising revenue through excessive advertisements .	Displays pop-up advertisements , redirects browsers, and reduces system performance.	YouTube or Web Browser where users may experience unwanted advertisements and browser redirections.
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Key Differences Between Malware Types:

- **Virus** requires user interaction and attaches itself to legitimate files.
- **Worm** spreads automatically through networks without any user action.
- **Ransomware** encrypts files and demands payment to restore access.
- **Trojan Horse** disguises itself as trusted software to deceive users.
- **Spyware** secretly monitors user activities and steals sensitive information.
- **Adware** mainly displays unwanted advertisements and redirects users to promotional websites.

TOPS TECH Task 1 (CS)

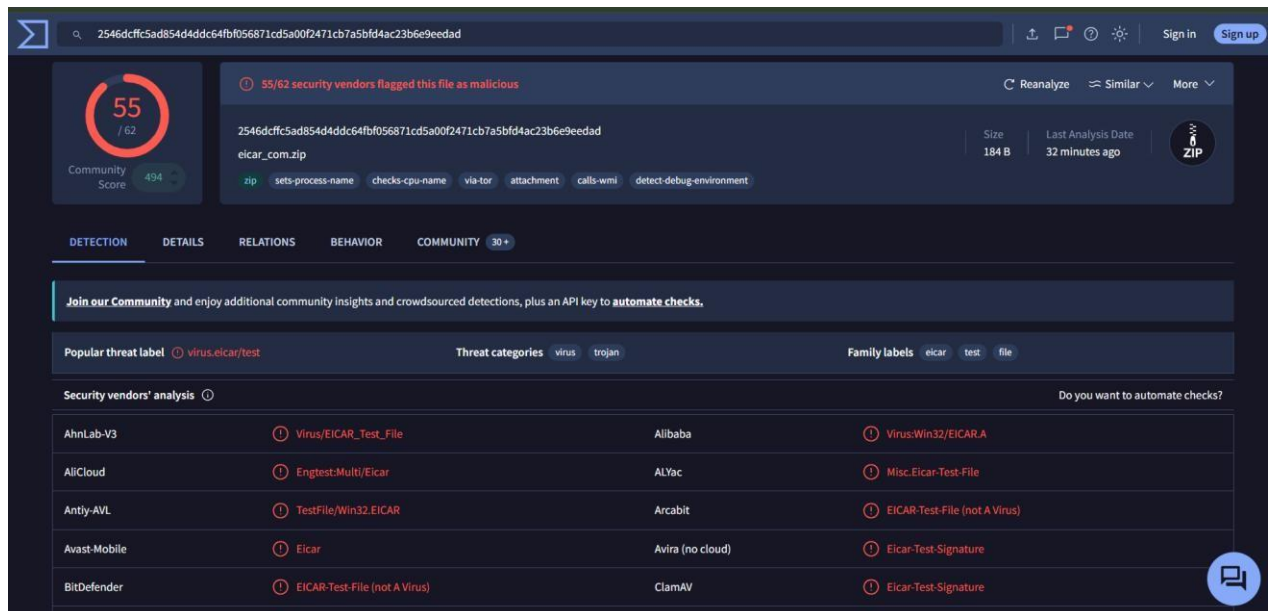
2. Download a harmless sample EICAR test file (not an actual virus) and upload it to VirusTotal. Take a screenshot of the detection results and write a short summary explaining which antivirus engines detected it and what labels they gave.

Ans:

Analyze the EICAR Test File Using VirusTotal

Objective

- The objective of this practical is to analyze the EICAR Anti-Malware Test File using VirusTotal and understand how multiple antivirus engines detect a harmless test file.
- The EICAR test file is not a real virus; it is a standardized test file created to verify whether antivirus software is functioning correctly.
- This practical helps in understanding malware detection techniques without exposing the system to actual malicious software.



The screenshot shows the VirusTotal analysis page for a file named 'eicar_com.zip' with SHA256 hash '2546dcffc5ad854d4ddc64fbf056871cd5a00f2471cb7a5bfd4ac23b6e9eedad'. The file size is 184 B and it was last analyzed 32 minutes ago. A community score of 55/62 is displayed, indicating that 55 out of 62 security vendors flagged the file as malicious. The file is categorized as a 'virus.eicar/test' and is associated with threat categories 'virus' and 'trojan'. The analysis shows that 30+ security vendors have detected the file. The table below lists the security vendors and their specific detections.

Security vendors' analysis	Detection
AhnLab-V3	Virus/EICAR_Test_File
Alibaba	Virus:Win32/EICAR.A
AliCloud	Engtest:Multi/Eicar
ALYac	Misc.Eicar-Test-File
Antiy-AVL	TestFile/Win32.EICAR
Arcabit	EICAR-Test-File (not A Virus)
Avast-Mobile	Eicar
Avira (no cloud)	Eicar-Test-Signature
BitDefender	EICAR-Test-File (not A Virus)
ClamAV	Eicar-Test-Signature

TOPS TECH Task 1 (CS)

3. Pick any free Android APK from a third-party site (not Google Play), upload it to VirusTotal, and analyze the report. List at least three suspicious indicators VirusTotal highlights, such as risky permissions or flagged behaviors. **Hint:** Focus on the 'Details' and 'Behavior' tabs in the VirusTotal report.

Ans:

Analyze an Android APK File Using VirusTotal

Objective

- The objective of this practical is to analyze an Android Application Package (APK) using VirusTotal and identify potential security risks.
- The analysis helps in understanding how antivirus engines and behavioral analysis tools detect suspicious permissions, risky activities, and malicious characteristics present in Android applications.

The screenshot displays the VirusTotal analysis interface for the file 'vlc-media-player-3.0.23-installer.exe'. The file's SHA-256 hash is 67592f28595d206dc41bb1cbe102e6a94b05a585d6b6de525f19428fb918fec. The community score is 12/70, with a red indicator showing it is malicious. The report is categorized under 'adware', 'downloader', and 'virus'. The 'Security vendors' analysis' section lists detections from various engines, including AhnLab-V3 (PUP/Win.Generic.R752265), DrWeb (Adware.Downware.20879), ESET-NOD32 (Win32/OffCore.E Potentially Unwanted...), K7GW (Adware (005693e61)), Malwarebytes (PUP.Optional.Softonic), CrowdStrike Falcon (Win/malicious_confidence_60% (D)), Elastic (Malicious (high Confidence)), K7AntiVirus (Adware (005693e61)), Kaspersky (Not-a-virus:HEUR:Downloader.Win32.So...), and Microsoft (PUADIManager:Win32/OffCore).

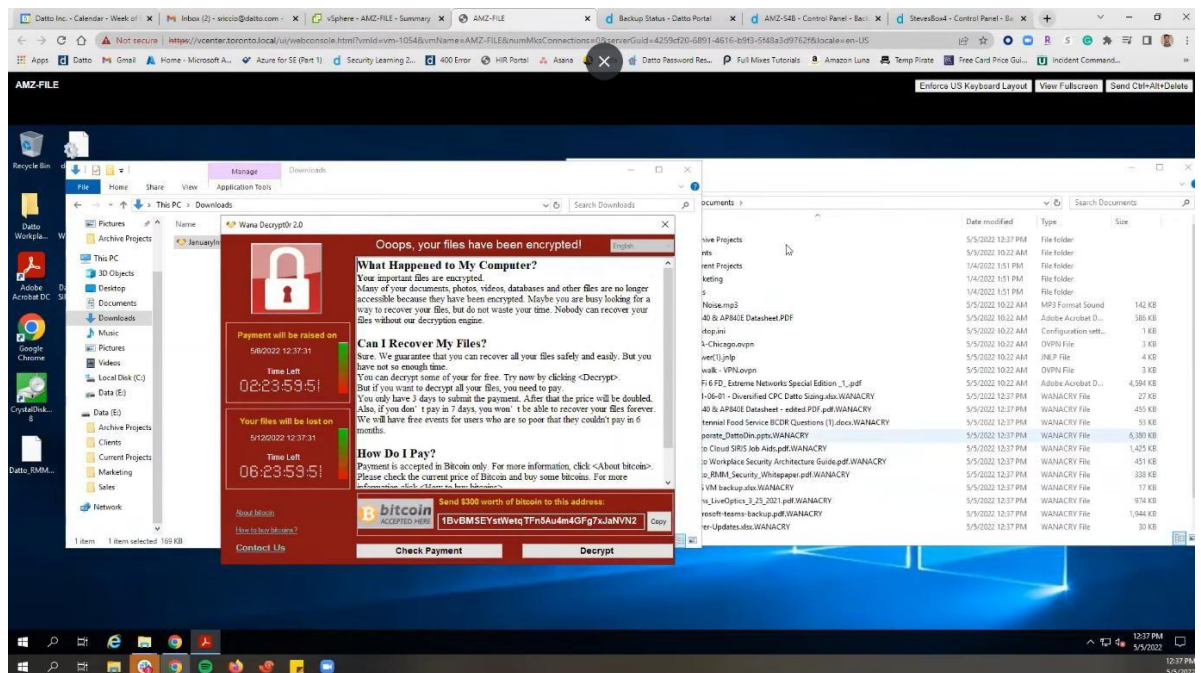
TOPS TECH Task 1 (CS)

4. Watch an online video demo (YouTube) of a ransomware attack in a sandboxed environment. Write a step-by-step summary of what happens to the victim's files and system, and explain how the sandbox prevents real damage.
- Hint:** Search for 'ransomware demo sandbox' on YouTube.

Ans:

Objective

- The objective of this practical is to understand the working of ransomware and observe its impact on a computer system in a secure sandboxed environment.
- A sandbox provides an isolated environment where malware can be executed safely without affecting the host operating system.
- This practical helps in understanding ransomware behavior, file encryption, and the importance of sandboxing in malware analysis.



TOPS TECH Task 1 (CS)

5. **Imagine you are building a Zomato-style food delivery app. List two ways malware (like spyware or adware) could affect your users, and suggest one security feature you would add to protect them.**

Ans:

Security Analysis for a Zomato-Style Food Delivery Application

Objective:

- The objective of this task is to understand how malware such as Spyware and Adware can affect users of a food delivery application.
- It also focuses on identifying an effective security feature that can be implemented to protect users' personal information, financial data, and overall application security.

Introduction:

- A food delivery application like Zomato or Swiggy stores a large amount of sensitive user information, including personal details, delivery addresses, mobile numbers, payment information, and location data.
- If malware infects a user's device or the application itself, attackers may steal confidential information, display unwanted advertisements, or redirect users to malicious websites.
- Therefore, implementing strong security measures is essential to ensure user privacy and maintain trust in the application.

How Spyware Can Affect Users:

- Spyware is malicious software designed to secretly monitor user activities and collect sensitive information without the user's knowledge.

Effect 1: Theft of Personal and Financial Information

- Spyware can capture usernames, passwords, credit/debit card details, UPI credentials, and login information entered into the food delivery application. Attackers may misuse this information to perform unauthorized transactions or gain access to user accounts.

TOPS TECH Task 1 (CS)

Effect 2: Tracking User Location and Personal Activities

- Food delivery applications require location access for order tracking and delivery services. Spyware can misuse this permission to continuously monitor the user's real-time location, travel history, and daily activities, leading to serious privacy violations.

How Adware Can Affect Users:

- Adware is software that displays unwanted advertisements or redirects users to promotional or malicious websites.

Effect 1: Excessive Pop-up Advertisements

- Adware may continuously display intrusive advertisements while users browse restaurants or place food orders. These advertisements degrade the user experience and reduce application performance.

Effect 2: Redirection to Fake or Malicious Websites

- Adware can redirect users to fraudulent websites that imitate payment pages or promotional offers. Users may unknowingly enter sensitive banking or personal information, resulting in financial fraud or identity theft.

Recommended Security Feature:

Multi-Factor Authentication (MFA)

- One of the most effective security features for a food delivery application is Multi-Factor Authentication (MFA).

MFA requires users to verify their identity using two or more authentication methods, such as:

- Password
- One-Time Password (OTP)
- Authentication Application
- Biometric Authentication (Fingerprint or Face Recognition)
- Even if an attacker obtains the user's password through spyware or phishing attacks, they cannot access the account without completing the second authentication step.

TOPS TECH Task 1 (CS)

Additional Security Measures:

To further improve the security of the application, the following measures should also be implemented:

- End-to-End Encryption for sensitive data transmission.
- Secure HTTPS communication for all network traffic.
- Strong password policy for user accounts.
- Regular security updates and vulnerability patching.
- Secure payment gateway integration.
- App integrity verification to detect tampering.
- Device-based login notifications.
- Automatic logout after periods of inactivity.
- Protection against rooted or jailbroken devices.
- Continuous monitoring for suspicious login activities.

Benefits of Implementing These Security Measures:

- Protects users' personal and financial information.
- Prevents unauthorized account access.
- Reduces the risk of phishing and malware attacks.
- Enhances user privacy and data confidentiality.
- Improves customer trust and application reliability.
- Ensures secure online payment transactions.
- Strengthens the overall cybersecurity of the application.

Result:

- The analysis shows that spyware and adware can significantly compromise the security and privacy of users of a food delivery application.
- Spyware may steal sensitive information and track user activities, while adware can display intrusive advertisements and redirect users to malicious websites.
- Implementing Multi-Factor Authentication along with other security mechanisms greatly reduces these risks and enhances the overall security of the application.